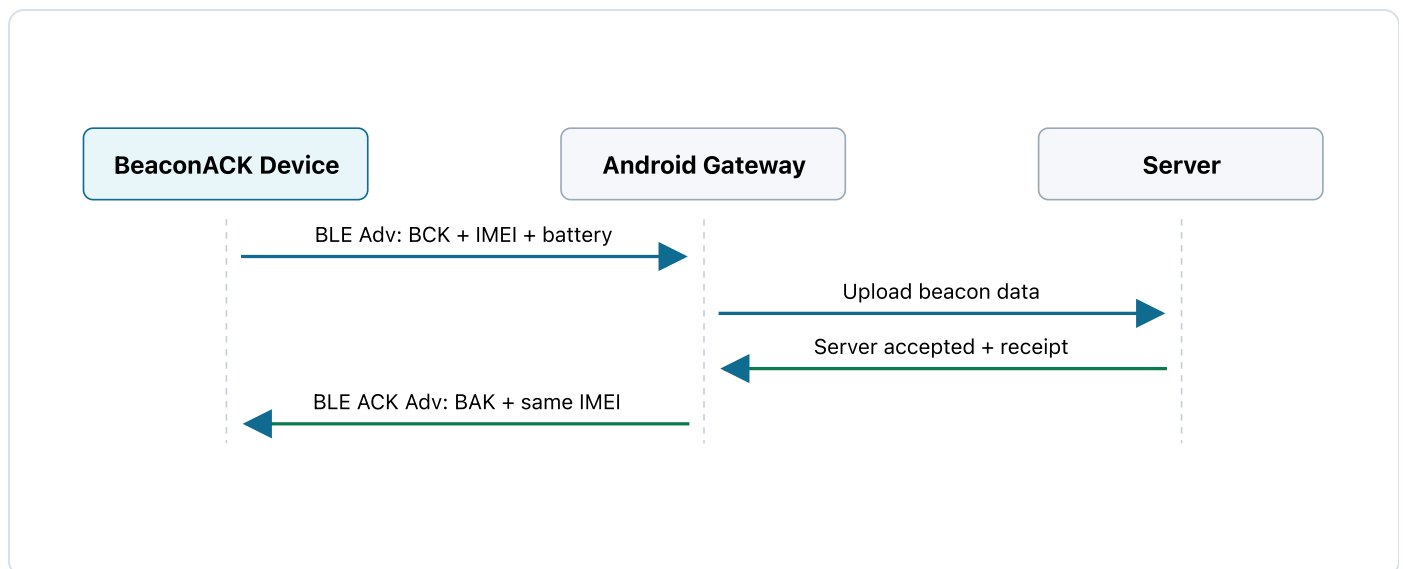


BeaconACK Android BLE Protocol

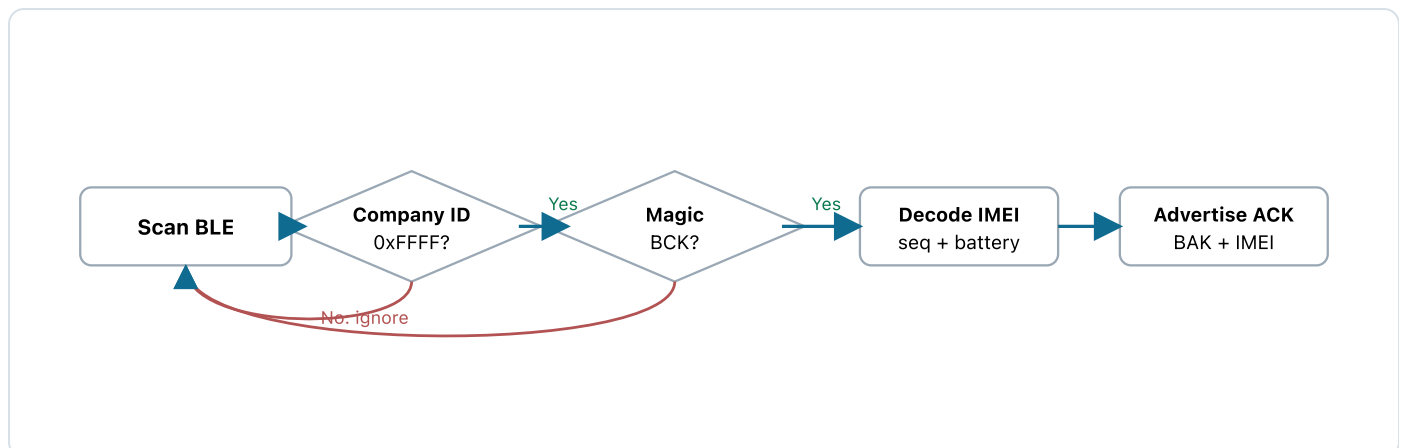
Protocol handoff for scanning BeaconACK BLE advertisements and sending connectionless ACK advertisements.

Android gateway responsibilities: scan manufacturer data for company ID 0xFFFF, detect BeaconACK packets by magic BCK, extract telemetry, upload it to the server, then advertise an ACK packet with magic BAK and the same IMEI.

Communication Flow



Gateway Decision Flow



Beacon Advertisement

Device scan response name:

```
BeaconACK
```

Do not rely only on the BLE name or MAC address. Identify the device by manufacturer data.

Full raw advertisement example:

```
02010614FFFFFFFF42434B01010079DF0D8648700000286810
```

Manufacturer Data

Bytes	Meaning
14	AD length: 20 bytes after this byte
FF	AD type: Manufacturer Specific Data
FF FF	Company ID: 0xFFFF, little-endian
42 43 4B	Magic: ASCII BCK
01	Protocol version
01 00	Beacon sequence, little-endian
79 DF 0D 86 48 70 00 00	IMEI uint64 LE
28	Battery percentage: $0x28 = 40\%$
68 10	Battery voltage: $0x1068 = 4200\text{ mV}$

Android ScanRecord Parsing

`Android ScanRecord.getManufacturerSpecificData(0xFFFF)` returns only the manufacturer payload after the company ID.

For the example above, Android receives:

```
42434B01010079DF0D8648700000286810
```

Offset	Size	Field	Expected / Encoding
0	3	Magic	ASCII BCK
3	1	Version	0x01
4	2	Sequence	uint16, little-endian
6	8	IMEI	uint64 little-endian
14	1	Battery percentage	uint8, 0 to 100
15	2	Battery voltage	uint16, little-endian, mV

Minimum valid Android manufacturer payload length: 17 bytes.

```
79 DF 0D 86 48 70 00 00 -> 123456789012345
```

Decode the 8 bytes as an unsigned little-endian integer, then format as decimal if the ASCII IMEI is needed.

Beacon Payload Layout

0..2	3	4..5	6..13	14	15..16
B C K	Version	Sequence LE	IMEI uint64 LE	Battery %	Battery mV LE

ACK Advertisement

Android should advertise a non-connectable legacy BLE packet named:

```
BeaconACK-ACK
```

Manufacturer company ID: 0xFFFF

Offset	Size	Field	Value / Encoding
0	3	Magic	ASCII BAK
3	1	Version	0x01
4	2	ACK sequence	uint16, little-endian. Can mirror beacon sequence.
6	8	IMEI	Same uint64 little-endian IMEI received from BeaconACK
14	1	Status	0x01 means sent to server / success
15	4	Receipt ID	uint32, little-endian

ACK manufacturer payload example for IMEI 123456789012345, sequence 1, status 1, receipt 0xA1B2C3D4:

```
42414B01010079DF0D864870000001D4C3B2A1
```

Full raw ACK advertisement equivalent:

```
02010616FFFFFFF42414B01010079DF0D864870000001D4C3B2A1
```

ACK Payload Layout

0..2	3	4..5	6..13	14	15..18
B A K	Version	ACK Sequence LE	Same IMEI uint64 LE	Status	Receipt ID LE

Timing

BeaconACK scans for ACK for 4 seconds after it sends its beacon.

Android should start the ACK advertisement immediately after parsing a valid BeaconACK packet and keep advertising for at least 1 second. Keeping it on for the full 4 second ACK window is safest during testing.

Implementation Notes

The Android gateway should use standard BLE scan APIs to read manufacturer data for company ID 0xFFFF.

For ACK, advertise the ACK manufacturer payload under the same company ID 0xFFFF. The ACK advertisement should be non-connectable and should include the local name BeaconACK-ACK.